

THREAT INTELLIGENCE REPORT



Modern Cybersecurity Threats: Analysis, Impact & Defense

Covering the 2024–2025 Global Threat Landscape

Mohamed Ibrahim Khan S M

Cybersecurity Intern

B.E. Computer Science & Engineering (Cyber Security)

Velammal Engineering College, Chennai

TABLE OF CONTENTS

1. Executive Summary	3
2. Introduction to Cybersecurity	4
2.1 What is Cybersecurity?	4
2.2 Why Cybersecurity Matters	4
2.3 Current Threat Landscape	4
3. Threat 1 — AI-Powered Phishing Attacks	5
4. Threat 2 — Ransomware-as-a-Service (RaaS)	6
5. Threat 3 — Cloud Security Misconfigurations	7
6. Threat 4 — IoT Vulnerabilities	8
7. Threat 5 — Zero-Day Exploits	9
8. Threat Summary Comparison	10
9. Conclusion & Future Scope	11

EXECUTIVE SUMMARY

EXECUTIVE SUMMARY

This Threat Intelligence Report, prepared as part of the Maincrafts Technology Cybersecurity Internship Program (Task-1), provides a comprehensive analysis of five critical cybersecurity threats dominating the 2024–2025 global threat landscape. The threats covered include AI-Powered Phishing Attacks, Ransomware-as-a-Service (RaaS), Cloud Security Misconfigurations, IoT Vulnerabilities, and Zero-Day Exploits. Each threat is examined through the lens of real-world incidents, impact analysis on individuals and organizations, and actionable defense recommendations. The findings of this report underscore a critical reality: the threat landscape is evolving at an unprecedented pace, driven by artificial intelligence, democratized cybercrime tools, and the expanding digital attack surface. Organizations and individuals alike must adopt a proactive, layered security posture to remain resilient.

KEY FINDINGS AT A GLANCE

5 Major threats analyzed with full impact and defense breakdown	5 Real-world case studies from 2017–2024 referenced	15+ Specific defense measures recommended across all threat categories	Critical AI-driven attacks represent the highest emerging threat vector in 2025
---	---	--	---

INTRODUCTION TO CYBERSECURITY

2.1 What is Cybersecurity?

Cybersecurity is the practice of protecting systems, networks, programs, devices, and data from digital attacks, unauthorized access, theft, and damage. It encompasses a broad set of technologies, processes, and practices designed to defend the confidentiality, integrity, and availability (CIA Triad) of information assets — whether stored, processed, or transmitted across digital infrastructure.

CIA TRIAD — THE FOUNDATION OF CYBERSECURITY

Confidentiality ensures that information is accessible only to those authorized to have access. Integrity guarantees that data is accurate, trustworthy, and has not been tampered with. Availability ensures that authorized users have reliable and timely access to information and systems when needed.

2.2 Why Cybersecurity Matters

In an era of accelerating digital transformation, cybersecurity is no longer optional — it is a fundamental business necessity and a personal imperative. The consequences of inadequate security span every domain:

For Individuals	Identity theft, financial fraud, loss of personal privacy, harassment, and account takeovers can devastate personal lives and finances.
For Businesses	Data breaches can result in regulatory fines (GDPR, India's DPDP Act), operational downtime, reputational damage, customer loss, and legal liability.
For Nations	Critical infrastructure such as power grids, hospitals, and financial systems are prime targets for nation-state threat actors pursuing strategic disruption.

2.3 Current Threat Landscape (2024–2025)

The cybersecurity landscape in 2024–2025 is characterized by the weaponization of artificial intelligence, the commoditization of cybercrime through dark-web service models, and an exponentially expanding attack surface driven by cloud adoption and IoT proliferation. According to IBM's Cost of a Data Breach Report 2024, the global average cost of a data breach reached **USD 4.88 million** — the highest ever recorded. Cybercrime damages are projected to exceed **USD 10.5 trillion annually by 2025**, surpassing the GDP of most nations. These figures demand immediate attention and a shift from reactive to proactive security postures.

THREAT 1 — AI-POWERED PHISHING ATTACKS

OVERVIEW

AI-Powered Phishing represents the most rapidly evolving social engineering threat of the modern era. Unlike traditional phishing, which relied on generic, easily identifiable mass emails, AI-driven attacks leverage large language models (LLMs), deepfake audio and video technology, and behavioral analysis to craft hyper-personalized, contextually convincing attacks at scale. Attackers use AI to scrape social media profiles, professional networks, and public data to generate spear-phishing emails that mimic the writing style of known contacts. Deepfake voice and video calls are used to impersonate executives in Business Email Compromise (BEC) and vishing attacks. In 2024, AI-generated phishing attacks increased by over 1,265% since the launch of ChatGPT, according to SlashNext's 2024 Phishing Report.

IMPACT ANALYSIS

IMPACT ON INDIVIDUALS

Individuals are tricked into revealing login credentials, banking details, and OTPs. Deepfake calls impersonating family members or employers can lead to direct financial wire transfers. Account takeovers result in identity theft, locked accounts, and long-term credit damage.

IMPACT ON ORGANIZATIONS

Organizations face BEC fraud losses averaging USD 50,000+ per incident. Successful phishing grants attackers initial access for larger intrusions. Regulatory penalties under GDPR and India's DPDP Act apply when customer data is exfiltrated. Reputational damage can erode client trust over years.

REAL-WORLD CASE STUDY

CASE STUDY: THE MGM RESORTS AI SOCIAL ENGINEERING ATTACK (2023)

In September 2023, the threat group Scattered Spider (UNC3944) compromised MGM Resorts International by executing a 10-minute vishing call to the IT helpdesk, impersonating an employee. The attackers had gathered personal information from LinkedIn. This single social engineering call led to a catastrophic ransomware deployment that disrupted MGM's hotel check-in systems, ATMs, and casino operations across Las Vegas. The breach resulted in an estimated USD 100 million in losses. This case illustrates how AI-enhanced social engineering can bypass technical defenses entirely by targeting the human element.

PREVENTIVE MEASURES

#	Defense Measure	Description	Priority
1	Multi-Factor Authentication (MFA)	Deploy FIDO2/hardware key-based MFA on all accounts to neutralize credential theft even when passwords are compromised.	Critical

2	Security Awareness Training	Conduct quarterly phishing simulation drills and AI-threat awareness programs for all employees, focusing on deepfake detection.	High
3	Email Authentication Protocols	Implement DMARC, DKIM, and SPF email authentication to prevent domain spoofing and email impersonation at the gateway level.	High
4	AI-Powered Email Filtering	Deploy intelligent email security gateways (e.g., Microsoft Defender for Office 365, Proofpoint) that use behavioral AI to detect and quarantine suspicious communications.	High

THREAT 2 — RANSOMWARE-AS-A-SERVICE (RAAS)

OVERVIEW

Ransomware-as-a-Service (RaaS) is a cybercrime business model in which ransomware developers lease their malicious software and infrastructure to affiliate attackers in exchange for a revenue share — typically 20–30% of ransom payments. This model has dramatically lowered the technical barrier to launching ransomware campaigns, transforming cybercrime into a scalable, subscription-based industry. RaaS platforms such as LockBit, BlackCat/ALPHV, and Cl0p operate like professional software companies — offering dashboards, customer support, negotiation portals, and even SLAs. In 2024, ransomware payments globally surpassed USD 1 billion for the first time, according to Chainalysis. Healthcare, education, and critical infrastructure sectors are the most heavily targeted.

IMPACT ANALYSIS

IMPACT ON INDIVIDUALS

Individuals can have personal files, photos, and documents permanently encrypted. Ransoms demanded from consumers typically range from USD 200 to USD 2,000. Victims who pay have no guarantee of data recovery. Personal data stolen prior to encryption may be published on dark-web leak sites.

IMPACT ON ORGANIZATIONS

Organizations face ransom demands ranging from hundreds of thousands to tens of millions of dollars. The average total cost of a ransomware attack including downtime, recovery, and reputational damage reached USD 4.54 million in 2024 (IBM). Critical operations can be halted for days or weeks. Regulatory non-compliance fines apply if customer data is exfiltrated.

REAL-WORLD CASE STUDY

CASE STUDY: WANNACRY RANSOMWARE (2017) — THE LARGEST RANSOMWARE ATTACK IN HISTORY

In May 2017, the WannaCry ransomware cryptoworm exploited EternalBlue — an NSA-developed exploit leaked by the Shadow Brokers — to propagate across unpatched Windows systems. Within 24 hours, WannaCry had infected over 230,000 systems across 150 countries. The UK's National Health Service (NHS) was among the most severely impacted — 80 out of 236 NHS trusts were affected, leading to the cancellation of approximately 19,000 medical appointments and an estimated GBP 92 million in damage. The attack is attributed to North Korea's Lazarus Group and demonstrated how a single unpatched vulnerability combined with a RaaS-style distribution model could cause global catastrophe.

PREVENTIVE MEASURES

#	Defense Measure	Description	Priority
1	Immutable Offline Backups	Maintain the 3-2-1 backup strategy: 3 copies of data, on 2 different media types, with 1 stored offline and air-gapped to prevent ransomware encryption of backups.	Critical

2	Endpoint Detection & Response (EDR)	Deploy EDR solutions (e.g., CrowdStrike Falcon, SentinelOne) capable of detecting and terminating ransomware behavior in real time before encryption completes.	Critical
3	Regular Patch Management	Establish a structured patch management cycle with a maximum 30-day window for critical vulnerability patching. Prioritize internet-facing systems and OS patches.	High
4	Network Segmentation	Isolate critical systems and limit lateral movement by segmenting networks using VLANs and Zero Trust micro-segmentation, preventing ransomware from propagating organization-wide.	High

THREAT 3 — CLOUD SECURITY MISCONFIGURATIONS

OVERVIEW

Cloud Security Misconfigurations occur when cloud infrastructure — spanning AWS, Microsoft Azure, and Google Cloud Platform (GCP) — is deployed without appropriate security controls, resulting in exposed storage buckets, excessive IAM permissions, publicly accessible databases, and unencrypted sensitive data. According to Gartner, through 2025, 99% of cloud security failures will be the customer's fault, driven by misconfiguration and mismanagement. The speed of cloud adoption, combined with skill gaps and the complexity of cloud-native services, makes misconfigurations the leading cause of cloud data breaches. The Shared Responsibility Model often leads organizations to incorrectly assume their cloud provider handles all security — a dangerous misconception that threat actors actively exploit.

IMPACT ANALYSIS

IMPACT ON INDIVIDUALS

Personal data including health records, financial information, and identity documents stored in misconfigured cloud services is exposed to unauthorized parties. Individuals may face identity theft or targeted fraud without even knowing their data was leaked. Downstream impacts include fraudulent loan applications and compromised tax records.

IMPACT ON ORGANIZATIONS

Organizations face GDPR fines of up to 4% of global annual turnover for data exposure. Intellectual property theft, exposure of unreleased products, and strategic plans can devastate competitive advantage. Extended breach detection times (average 277 days) amplify damage. The reputational cost of a public breach disclosure can suppress stock value and erode customer trust for years.

REAL-WORLD CASE STUDY

CASE STUDY: CAPITAL ONE CLOUD MISCONFIGURATION BREACH (2019)

In July 2019, a former Amazon Web Services engineer exploited a misconfigured Web Application Firewall (WAF) on Capital One's AWS infrastructure to access over 100 million US and Canadian customer records — including Social Security numbers, bank account numbers, credit scores, and transaction history. The attacker used a Server-Side Request Forgery (SSRF) vulnerability combined with an overly permissive IAM role to exfiltrate data from an S3 bucket. Capital One faced an USD 80 million OCC regulatory fine and over USD 190 million in class-action lawsuit settlements. This case established the critical importance of least-privilege IAM policies and SSRF protection in cloud environments.

PREVENTIVE MEASURES

#	Defense Measure	Description	Priority
---	-----------------	-------------	----------

1	Cloud Security Posture Management (CSPM)	Deploy automated CSPM tools (e.g., Prisma Cloud, AWS Security Hub) to continuously scan and remediate misconfigurations across cloud environments in real time.	Critical
2	Least Privilege IAM Policy	Enforce the principle of least privilege for all IAM roles and service accounts. Conduct quarterly access reviews and remove unused permissions and orphaned accounts.	High
3	Cloud Encryption & Key Management	Encrypt all data at rest and in transit using cloud-native encryption services. Manage encryption keys through a dedicated KMS (Key Management Service) with audit logging.	High
4	Infrastructure as Code (IaC) Security Scanning	Integrate security scanning into CI/CD pipelines using tools like Checkov or Terrascan to detect misconfigurations before deployment to production environments.	Medium

THREAT 4 — IOT VULNERABILITIES

OVERVIEW

The Internet of Things (IoT) encompasses billions of internet-connected devices — from smart home assistants, security cameras, and industrial sensors to medical devices and smart city infrastructure. By 2025, there are estimated to be over 75 billion IoT devices globally, yet the majority are deployed without adequate security controls. IoT devices typically suffer from hardcoded default credentials, unencrypted communications, absent firmware update mechanisms, and minimal security validation. These weaknesses make them prime targets for botnet recruitment (e.g., Mirai), lateral movement into enterprise networks, and industrial sabotage. The convergence of IT and Operational Technology (OT) networks in Industry 4.0 environments has elevated IoT vulnerabilities from nuisances to critical infrastructure risks.

IMPACT ANALYSIS

IMPACT ON INDIVIDUALS

Smart home devices such as IP cameras and baby monitors can be hijacked for surveillance or blackmail. Compromised smart locks and connected vehicles pose direct physical safety risks. Personal data collected by wearables and voice assistants is at risk of exposure through insecure cloud backends.

IMPACT ON ORGANIZATIONS

Industrial IoT (IIoT) attacks on manufacturing, energy, and utilities can cause physical damage, production shutdown, and safety hazards. Healthcare IoT — including insulin pumps and pacemakers — represents life-critical attack surfaces. A single compromised IoT device on a corporate network can serve as an entry point for broader intrusion campaigns. Incident response in OT environments is significantly more complex and costly than in traditional IT.

REAL-WORLD CASE STUDY

CASE STUDY: MIRAI BOTNET DDOS ATTACK (2016)

In October 2016, the Mirai botnet — composed of hundreds of thousands of hijacked IoT devices including CCTV cameras and DVRs secured only with default credentials — was weaponized to launch a record-breaking 1.2 Tbps Distributed Denial of Service (DDoS) attack against Dyn, a major DNS provider. The attack brought down major portions of the internet across the US East Coast, rendering services including Twitter, Netflix, Reddit, GitHub, and PayPal inaccessible for hours. The Mirai source code was subsequently released publicly, spawning hundreds of derivative botnets. This event fundamentally changed how the security community views IoT device hardening and demonstrated that consumer devices could be weaponized into cyber weapons of mass disruption.

PREVENTIVE MEASURES

#	Defense Measure	Description	Priority
---	-----------------	-------------	----------

1	Default Credential Elimination	Enforce mandatory password change during device setup. Disable default accounts and implement device-specific unique credentials from the factory or provisioning stage.	Critical
2	Network Segmentation & IoT VLANs	Place all IoT devices on isolated network segments (VLANs) with strict firewall rules preventing lateral movement into corporate or sensitive networks.	High
3	Firmware Update & Patch Management	Implement automated over-the-air (OTA) firmware updates for IoT devices. Maintain an asset inventory and monitor CVEs specific to deployed device models.	High
4	IoT Security Standards Compliance	Adhere to NIST SP 800-213, ETSI EN 303 645, and emerging EU Cyber Resilience Act requirements for IoT device security baselines and manufacturer accountability.	Medium

THREAT 5 — ZERO-DAY EXPLOITS

OVERVIEW

A Zero-Day Exploit is a cyberattack that targets a previously unknown software vulnerability for which no patch or official fix exists at the time of exploitation. The term 'zero-day' refers to the zero days of warning the software vendor has had to address the flaw. Zero-days represent the most sophisticated and dangerous category of cyber threat because traditional signature-based defenses are ineffective against them — there is simply no known indicator of compromise (IOC) to detect. These vulnerabilities are discovered by security researchers, nation-state intelligence agencies, and criminal actors. A single zero-day exploit in a widely used product can compromise millions of systems globally before a patch is available. In 2024, over 70 zero-day vulnerabilities were actively exploited in the wild, according to Google's Project Zero — the highest annual count recorded.

IMPACT ANALYSIS

IMPACT ON INDIVIDUALS

Individuals using vulnerable browsers, mobile OS versions, or productivity software can be silently compromised without any interaction beyond visiting a malicious website (drive-by downloads). Spyware such as Pegasus, delivered via zero-day exploits, has been used to surveil journalists, activists, and private individuals. Full device takeover enables theft of messages, calls, photos, and location data.

IMPACT ON ORGANIZATIONS

Organizations face silent, undetectable intrusions that establish persistent access weeks or months before detection. Supply chain zero-days can simultaneously compromise all downstream customers. Financial institutions, government agencies, and defense contractors are primary targets. The average dwell time before detection of a zero-day-initiated breach exceeds 200 days, enabling extensive data exfiltration.

REAL-WORLD CASE STUDY

CASE STUDY: SOLARWINDS SUPPLY CHAIN ZERO-DAY ATTACK (2020)

The SolarWinds attack, discovered in December 2020, represents one of the most sophisticated and consequential supply chain compromises in history. Russian state-sponsored threat actors (APT29 / Cozy Bear) inserted malicious code into a software update for SolarWinds' Orion IT monitoring platform — used by 33,000+ organizations globally. The trojanized update was digitally signed and distributed through official channels, bypassing all conventional security controls. Over 18,000 organizations installed the compromised update, including the US Treasury, Department of Homeland Security, Microsoft, Intel, and Cisco. The attackers maintained covert access for 9 months before detection. The attack demonstrated how zero-day techniques embedded in trusted supply chain software can achieve unprecedented scale and stealth.

PREVENTIVE MEASURES

#	Defense Measure	Description	Priority
---	-----------------	-------------	----------

1	Patch Management & Vulnerability Prioritization	Implement a risk-based patch management framework. Subscribe to CISA KEV (Known Exploited Vulnerabilities Catalog) and prioritize patches for actively exploited CVEs within 24–72 hours.	Critical
2	Web Application Firewall (WAF) & Virtual Patching	Deploy WAF solutions with virtual patching capability to block exploitation attempts targeting unpatched vulnerabilities while official patches are developed and tested.	High
3	Behavioral Detection & Threat Hunting	Deploy behavior-based detection (UEBA, EDR) and conduct proactive threat hunting to identify anomalous activity indicative of zero-day exploitation even without known signatures.	High
4	Software Supply Chain Security	Implement SBOM (Software Bill of Materials) for all deployed software. Verify code signing, monitor for anomalous update behavior, and apply Zero Trust principles to software supply chain validation.	High

THREAT SUMMARY COMPARISON

The following matrix provides a consolidated view of all five threats analyzed in this report, enabling rapid comparison of severity, primary targets, and priority defense recommendations.

Threat	Category	Severity	Primary Target	Key Defense
AI-Powered Phishing	Social Engineering	CRITICAL	Individuals & Employees	MFA + Security Training
Ransomware-as-a-Service	Malware / Extortion	CRITICAL	Organizations	Offline Backups + EDR
Cloud Misconfigurations	Infrastructure Risk	HIGH	Organizations / Cloud Assets	CSPM + IAM Hardening
IoT Vulnerabilities	Device Security	HIGH	Consumers & Industries	Network Segmentation
Zero-Day Exploits	Vulnerability Exploit	CRITICAL	All Systems	Patch Management + WAF

THREAT RISK MATRIX

Threat	Likelihood (1–5)	Impact (1–5)	Risk Level
AI-Powered Phishing	●●●●●	●●●●●	CRITICAL
Ransomware-as-a-Service	●●●●●	●●●●●	CRITICAL
Cloud Misconfigurations	●●●●●	●●●●●	CRITICAL
IoT Vulnerabilities	●●●●●	●●●●●	HIGH
Zero-Day Exploits	●●●●●	●●●●●	CRITICAL

CONCLUSION & FUTURE SCOPE

9.1 Conclusion

This Threat Intelligence Report has examined five of the most critical cybersecurity threats defining the 2024–2025 global security landscape: AI-Powered Phishing, Ransomware-as-a-Service, Cloud Security Misconfigurations, IoT Vulnerabilities, and Zero-Day Exploits. The analysis reveals a consistent, sobering theme — the threat landscape is not merely growing in volume, but in sophistication, automation, and accessibility to adversaries of all skill levels.

Proactive cybersecurity is no longer a choice reserved for large enterprises. Every organization — regardless of size, sector, or geography — is a potential target. The convergence of AI-driven attack automation, ransomware commoditization, and cloud and IoT expansion has eliminated traditional security perimeters. The only defensible posture in this environment is one built on continuous monitoring, layered defenses, Zero Trust architecture, and a culture of security awareness extending from the boardroom to the end user.

CORE TAKEAWAY

The most expensive security investment an organization can make is reacting after a breach. A USD 10,000 investment in proactive security today can prevent a USD 4.88 million breach tomorrow. Cybersecurity is not a cost center — it is a business continuity imperative.

9.2 Future Scope & Emerging Threats

As this report is finalized, several emerging threat vectors demand vigilance from cybersecurity professionals in the coming years:

Quantum Computing Threats: Quantum computers will eventually break RSA and ECC encryption. Organizations must begin transitioning to NIST-approved post-quantum cryptographic standards (CRYSTALS-Kyber, CRYSTALS-Dilithium) now.

AI-vs-AI Warfare: Security tools will increasingly rely on AI for defense, while attackers will simultaneously use AI to probe, evade, and adapt. This arms race will define the next decade of cybersecurity.

Deepfake-Enabled Financial Fraud: AI-generated deepfake video calls impersonating executives for wire transfer authorization (CFO fraud) represent an imminent and growing threat to financial operations.

5G and Edge Computing Risks: The widespread rollout of 5G networks and edge computing introduces new attack surfaces, particularly for critical infrastructure and autonomous vehicle systems.

Regulatory Evolution: Frameworks such as India's DPDP Act, the EU AI Act, and evolving SEC cybersecurity disclosure rules will significantly reshape organizational security obligations.

9.3 Commitment to Continuous Learning

Cybersecurity is a discipline where standing still is equivalent to falling behind. The threats documented in this report will evolve — new variants will emerge, old defenses will be circumvented, and the attack surface

will continue to expand. The fundamental response to this reality is an unwavering commitment to continuous learning: following threat intelligence feeds such as CISA KEV and MITRE ATT&CK; pursuing professional certifications (ISC2 CC, CompTIA Security+, CEH, OSCP), and engaging with the global security community through platforms such as HackTheBox, TryHackMe, and CyberDefenders.

This report was prepared with diligence, professional integrity, and a commitment to advancing cybersecurity awareness. I am grateful for the opportunity provided by Maincrafts Technology and look forward to applying these learnings in real-world security operations.

Mohamed Ibrahim Khan S M

Cybersecurity Intern | Maincrafts Technology

May 17, 2026

Task-1 | Threat Intelligence Report